Kindly upload your project work in Zipped or PDF file

Link: https://forms.gle/6KLAYwB5FXsDsfmy6

Practical Assignment : House Price Prediction Using MLOps

1. Problem statement

Question:

A real estate company wants to develop a Machine Learning application that predicts the price of a house based on its size, number of bedrooms, number of bathrooms and age of the property.

Your task is to develop a simple Machine Learning model and implement an MLOps workflow to train, evaluate, track and save the model.

2. Dataset requirements

Create a dataset named house_prices.csv with the following columns:

Column	Description
area_sqft	House area in square feet
bedrooms	Number of bedrooms
bathrooms	Number of bathrooms
house_age	Age of house in years
price	House price (target)

Example dataset:

Area	Bedrooms	Bathrooms	Age	Price (₹)
1000	2	1	5	35,00,000
1500	3	2	3	55,00,000
2000	3	2	8	65,00,000
2500	4	3	2	95,00,000
1200	2	2	10	38,00,000

These are illustrative values for the exercise.

3. Practical tasks

Ask students to complete the following activities independently.

Task 1: Dataset preparation

- Create house_prices.csv.
- Load the dataset using Pandas.
- Display the first five records.
- Check missing values.

Task 2: Data preprocessing

- Select independent variables (X).
- Select dependent variable (y).
- Split data into 80% training and 20% testing.

Task 3: Model development

- Train a Linear Regression model.
- Generate house price predictions.
- Calculate MAE, RMSE and R^2.

Task 4: Experiment tracking

- Create an MLflow experiment named House Price Prediction.
- Record model parameters.
- Record evaluation metrics.
- Save the trained model artifact.

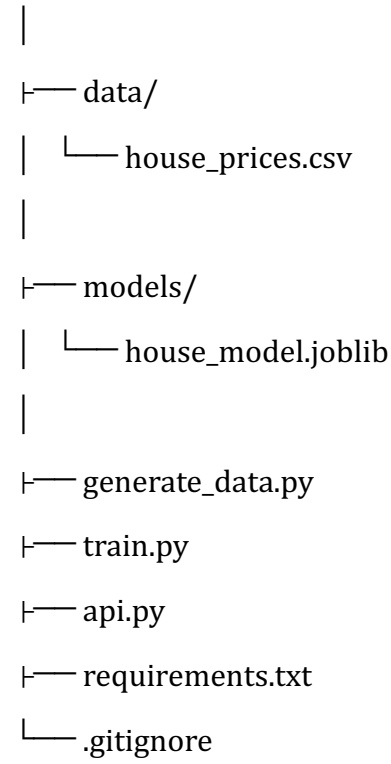
Task 5: Model versioning

- Save the model using Joblib.
- Maintain the Python source code using Git.
- Record the dependencies in requirements.txt.

Task 6: Model deployment

- Create a simple FastAPI application.

- Accept house details as input.
- Return the predicted house price.
- Test using Swagger UI.

4. Expected project structure

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